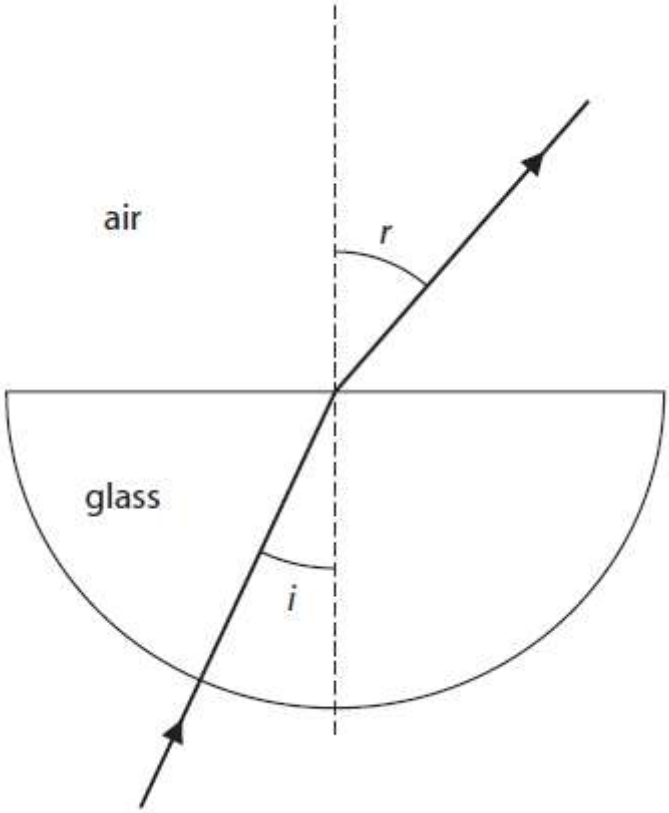
Questions

Q1.

A student investigates the way light passes through glass.
The diagram shows the path of a ray of light through the glass.



(a) State the scientific name for the dotted line in the diagram.

(1)

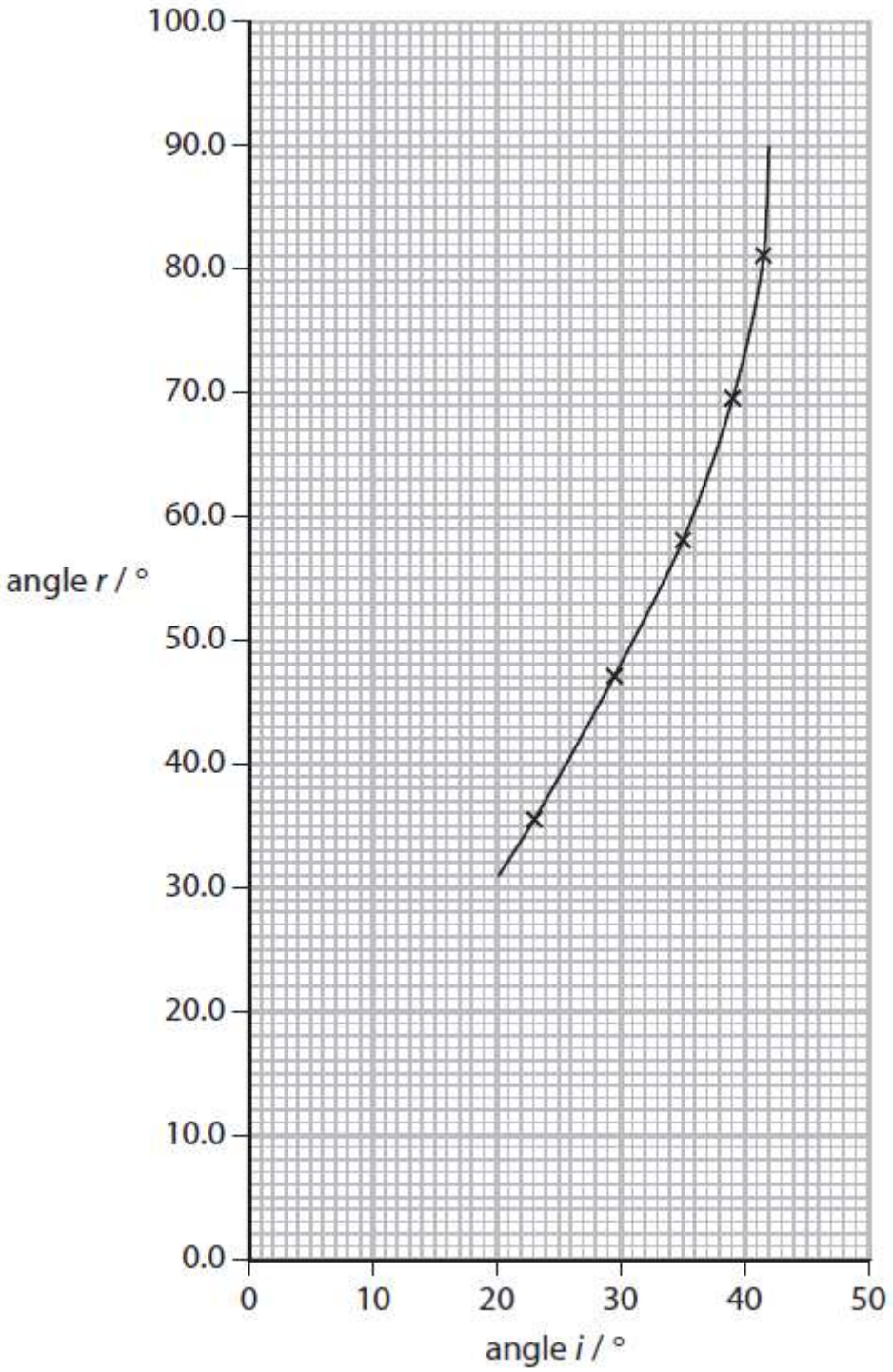
.....

(b) The student measures several values of angle *i* and angle *r*.
She plots some of her results on the graph.
The table shows results that she has not plotted.

angle *i*	angle *r*
0°	0°
6°	9°

(i) Plot these results on the graph.

(2)



(ii) Continue the line on the graph through the results you have plotted.

(1)

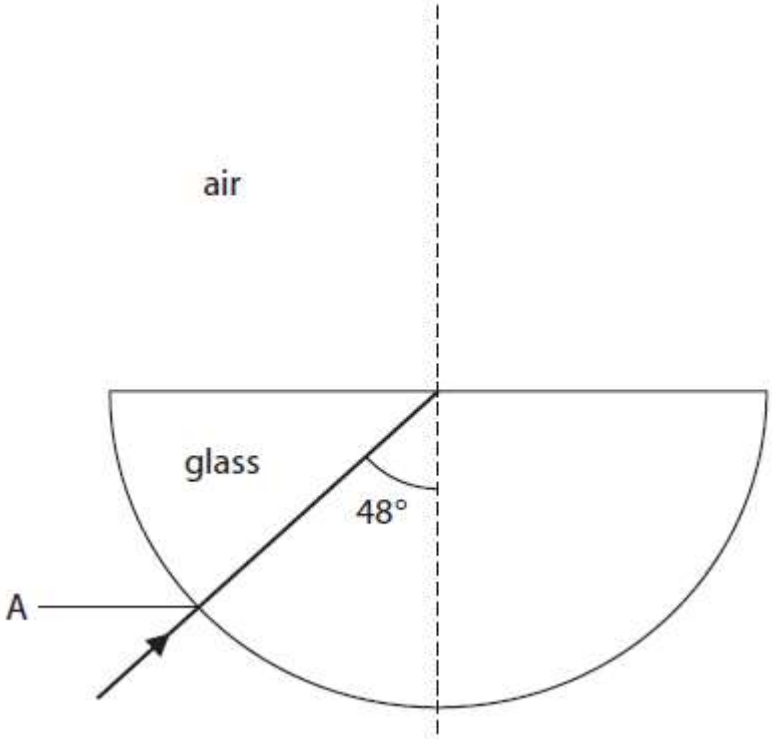
(iii) Write down the value of angle *i* when angle *r* = 90°.

(1)

angle *i* = ... °

(c) (i) Complete the diagram to show what happens to the ray of light when angle *i* is 48°.

(2)



(ii) State why the ray of light does not change direction when it enters the glass at A.

(1)

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.....

(Total for Question = 8 marks)

Q2.

Figure 3 shows a ray of light going from air to glass.

Fill in the labels in Figure 3 using words from the box.

critical	incident	normal	reflected	refracted

(3)

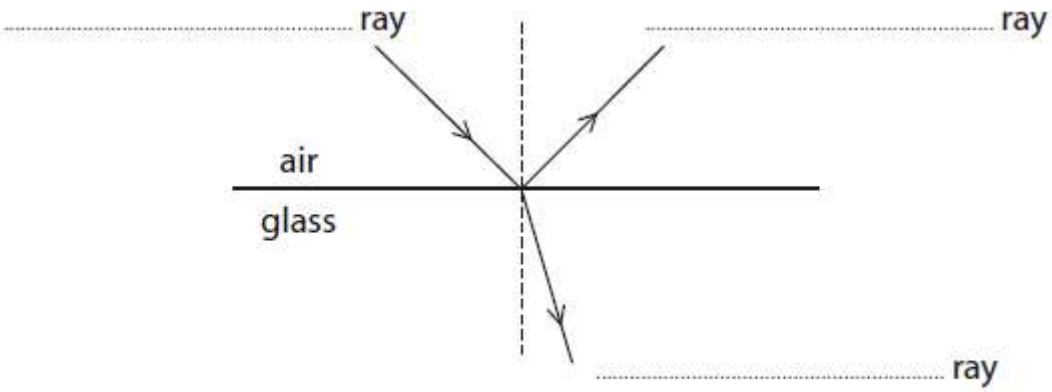

Figure 3

(Total for question = 3 marks)

Q3.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which of these always increases as a sound gets louder?

(1)

☐ **A** amplitude
☐ **B** frequency
☐ **C** speed
☐ **D** wavelength

(Total for question = 1 mark)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 7 shows a wave on the surface of water.

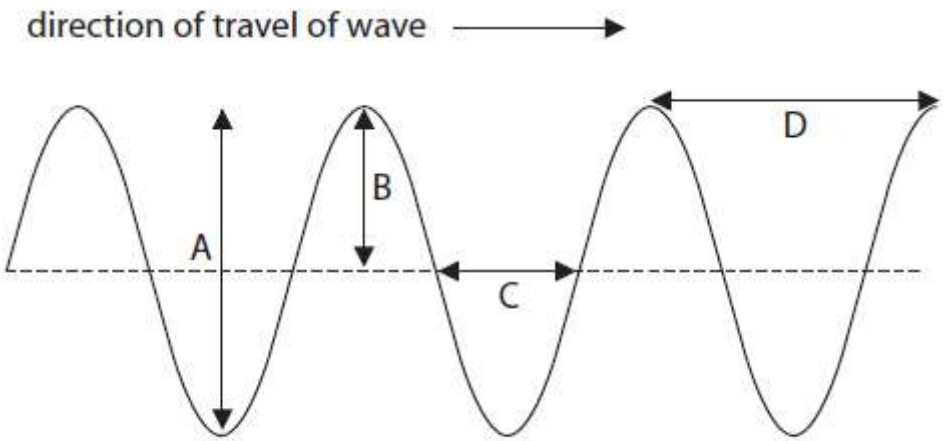

Figure 7

(i) Which of the arrowed lines shows the amplitude of the wave?

(1)

☐ **A**
☐ **B**
☐ **C**
☐ **D**

(ii) Explain why the wave shown in Figure 7 is a transverse wave.

(2)

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.....

(Total for question = 3 marks)

Q5.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Microwave ovens use microwaves.

Figure 1 shows a slice of bread that has been in a microwave oven.

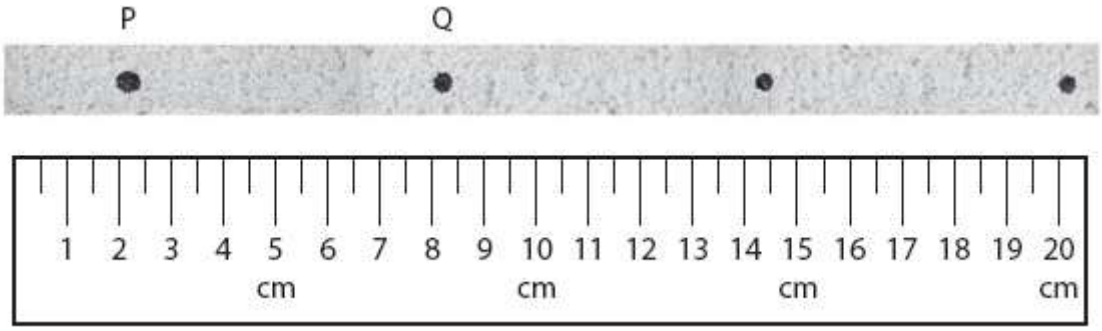

Figure 1

The dark spots are caused by the microwaves.

The spots are half a wavelength apart.

A technician measures the distance between P and Q to determine the wavelength of the microwaves.

(i) Which one of these is the wavelength measured using this method?

(1)

☐ **A** 2 cm

☐ **B** 6 cm

☐ **C** 12 cm

☐ **D** 20 cm

(ii) Describe how the technician could obtain a more accurate value for the wavelength from Figure 1.

(2)

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.....

.....

(Total for question = 3 marks)

Q6.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Which one of these is a longitudinal wave?

(1)

☐ **A** gamma

☐ **B** infrared

☐ **C** radio

☐ **D** sound

(Total for question = 1 mark)

Q7.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A sound wave can transfer information across a room.

Which row of the table shows what else a sound wave can transfer?

(1)

	can transfer energy	can transfer air
☐ A	yes	yes
☐ B	yes	no
☐ C	no	yes
☐ D	no	no

(Total for question = 1 mark)

Q8.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Figure 4 shows a wave on the sea moving towards a beach.

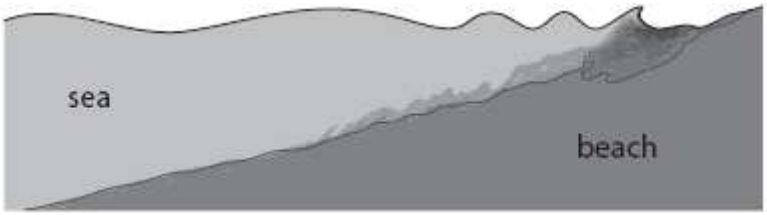

Figure 4

Which one of these is correct?

(1)

☐ **A** amplitude is the distance between two waves

☐ **B** frequency is the number of waves arriving at the beach per second

☐ **C** speed is the number of waves arriving at the beach per second

☐ **D** wavelength is the height of a wave

(Total for question = 1 mark)

Q9.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Figure 10 shows a water wave.

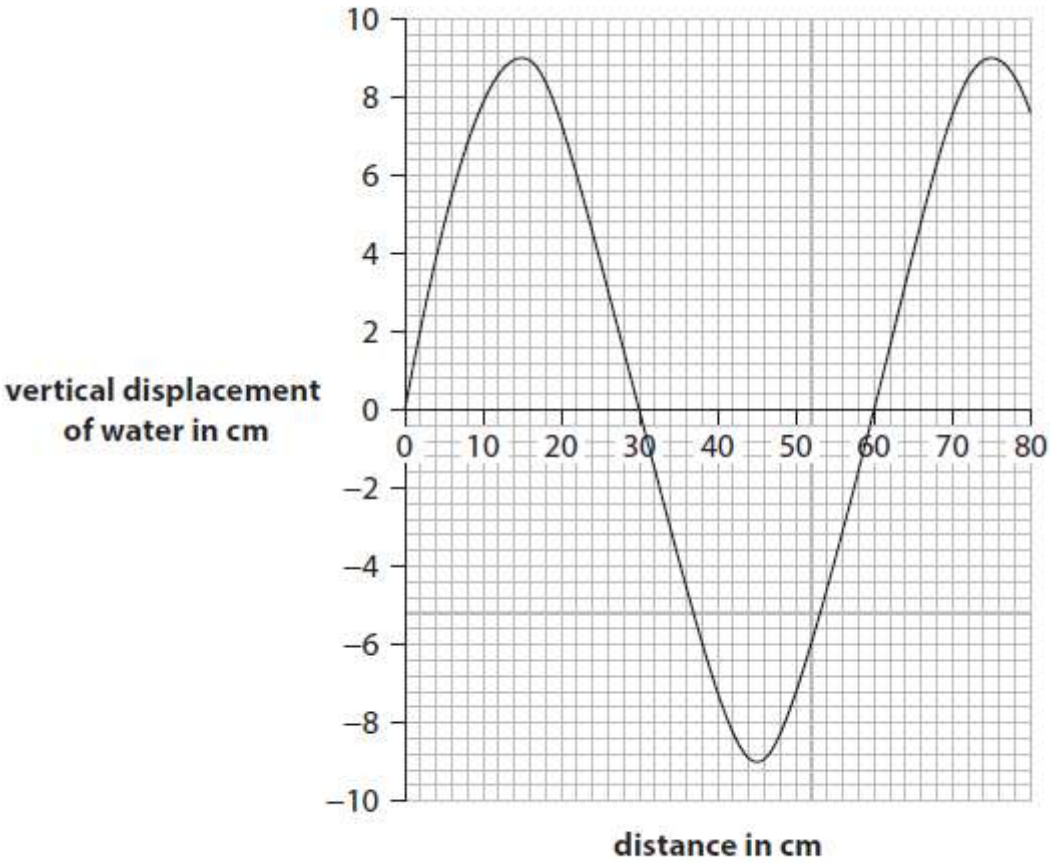

Figure 10

Which of these is the amplitude of the wave shown in Figure 10?

(1)

- ☒ **A** 9 cm
- ☒ **B** 18 cm
- ☒ **C** 30 cm
- ☒ **D** 60 cm

(Total for question = 1 mark)

Q10.

A man throws a stone into a pond.

Devise a method of measuring the frequency of the waves.

(2)

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(Total for question = 2 marks)

Q11.

Figure 2 shows water waves spreading out from a source.

A student measures the wavelength of the waves.

He uses a ruler to measure the distance from one crest to the next crest.

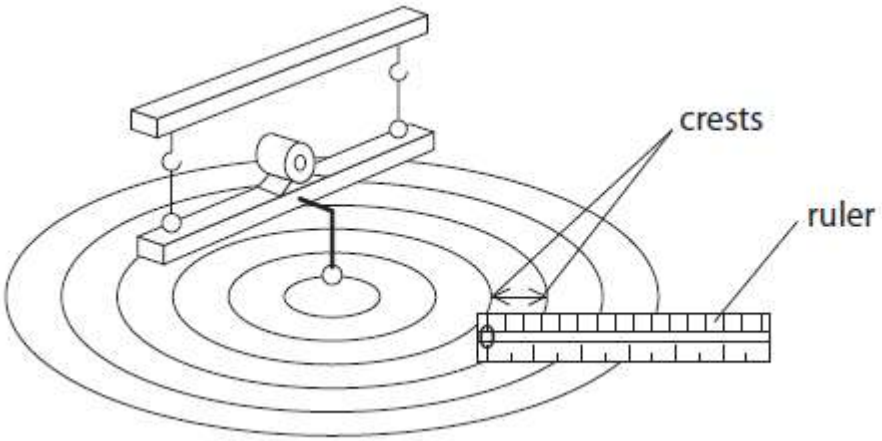

Figure 2

Explain how to improve the student's method for measuring the wavelength.

(2)

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.....

.....

(Total for question = 2 marks)

Mark Scheme

Q1.

Question Number	Answer	Acceptable answers	Mark
(a)	normal (1)	normal line	(1)

Question Number	Answer	Acceptable answers	Mark
(b)(i)	plot the points: • 0,0 (1) • 6,9 (1)	allow within one square tolerance. Bod if 0,0 not clearly visible but must be able to see a plotted point for 6,9 If they plot more than 2 points, take a mark off for each incorrect one plotted.	(2)

Question Number	Answer	Acceptable answers	Mark
(b)(ii)	straight line through both points joining existing curve (1)	Reject multiple lines and unreasonably wavering lines. allow ecf from wrongly plotted points, including curves if plausible	(1)

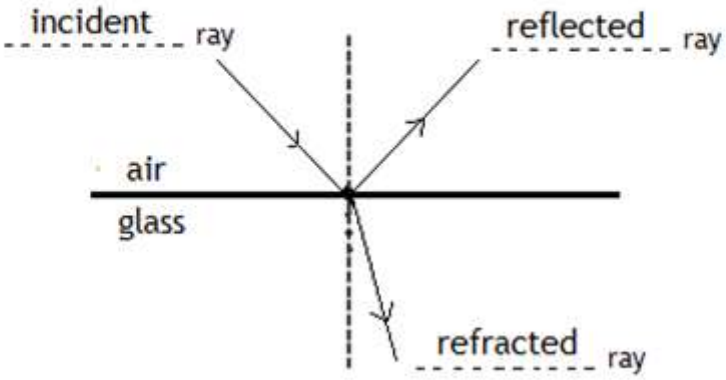
Question Number	Answer	Acceptable answers	Mark
(b)(iii)	42° (1) +/- 0.5°		(1)

Question Number	Answer	Acceptable answers	Mark
(c)(i)	diagram showing: • reflection (1) • angle of incidence = angle of reflection (1)	reject (for this marking point) with an additional partial refraction / ray along boundary judge by eye allow angles marked as equal	(2)

Question Number	Answer	Acceptable answers	Mark
(c)(ii)	The idea that it enters along the normal	At 90° to the surface / at right angles to the surface / along a radius / perpendicular to the tangent / hits straight on reject 'goes through centre of glass'	(1)

(Total for Question = 8 marks)

Q2.

Question Number:	Answer	Additional guidance	Mark
		All 3 labels correct 3 marks 2 labels correct 2 marks 1 label correct 1 mark	**(3)**

Q3.

Question number	Answer	Additional guidance	Mark
	A amplitude B, C and D are incorrect because they are independent of intensity		**1** **AO1.1**

Q4.

Question Number	Answer	Mark
(i)	**B** the line shows the amplitude **A** is incorrect the line shows twice the amplitude **C** is incorrect the line shows half the wavelength **D** is incorrect the line shows the wavelength	**(1)** **AO1**

Question Number	Answer	Additional guidance	Mark
(ii)	an explanation linking vibration/oscillation (1) perpendicular / at right angles / 90° (to the direction of travel of the wave/direction of energy transfer) (1)	 accept up and down	(2) AO1

Q5.

Question number	Answer	Additional guidance	Marks
(i)	C 12 cm		(1)

Question number	Answer	Additional guidance	Marks
(ii)	An answer that combines the following points of understanding to provide a logical description: • measure across several spots (1) • obtains the wavelength e.g. by dividing (the measurement) by the appropriate number (1)	 allow idea of measurement in mm / use a ruler with millimetre divisions allow 'put the ruler closer to the bread' ignore 'repeat the test'	(2)

Q6.

Question number	Answer	Additional guidance	Marks
	D sound		(1)

Q7.

Question number	Answer	Additional guidance	Mark
	B A, C and D are incorrect because energy is transferred but air is not transferred		**1** **AO1.1**

Q8.

Question number	Answer	Additional guidance	Marks
	B frequency is the number of waves arriving at the beach per second		(1)

Q9.

Question number	Answer	Additional guidance	Mark
	A 9cm B is incorrect because amplitude is measured from zero to the peak displacement C is incorrect because this is half the wavelength D is incorrect because this is the wavelength.		**1** **AO2.1**

Q10.

Question number	Answer	Additional guidance	Mark
	An answer that combines the following points to provide a method: • use a stop watch (1) • count number of waves that reach the bank in a given time (1)		(2)

Q11.

Question Number	Answer	Additional guidance	Mark
	an explanation linking:		(2)
	• measure across more than one (wavelength) (1)	use a more accurate device (finer divisions) use a camera / picture/strobe(light) (so the waves are not moving)	AO 3 3b
	• divide by the number of wavelengths (1)	count the number of wavelengths must be talking about **measuring**, NOT changing the wavelength etc.	